

Multiple integrals – 5

Asymptotics of multiple integrals

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Theoretical minimum.

Theorem 1 (Laplace’s method in $\mathbb{R}^n$). If S attains its maximum on a domain D at a single interior point x_0 with $\det S''(x_0) \neq 0$, then

$$\int_D f(x) e^{\lambda S(x)} dx \sim f(x_0) e^{\lambda S(x_0)} \left(\frac{2\pi}{\lambda} \right)^{n/2} \frac{1}{\sqrt{|\det S''(x_0)|}}, \quad \lambda \rightarrow +\infty.$$

The one-dimensional case is in Limits – 6; the multidimensional proof is the same after diagonalising the Hessian.

Theorem 2 (the maximum on the boundary). If the maximum of S is attained at a boundary point with $\nabla S \neq 0$, the answer changes: the contribution is of order λ^{-1} per transversal direction instead of $\lambda^{-1/2}$, so the whole integral is of order $\lambda^{-(n+1)/2}$ up to the geometry of the boundary. Integrals of the form $n^k \iint x^n y^n f$ are of exactly this kind: the mass concentrates at the corner of the square.

Fact 1 (approximate identities in several variables). A family of non-negative kernels with total mass one concentrating at a point reproduces the value of a continuous function there; in several variables the standard examples are the Gaussian $\left(\frac{n}{\pi}\right)^{n/2} e^{-n\|x\|^2}$ and the products of one-dimensional Landau kernels. This is the mechanism behind every limit of the form $c_n \iint K_n f \rightarrow f(x_0)$.

Fact 2 (concentration). On the sphere S^{n-1} and on the ball in $\mathbb{R}^n$ the measure concentrates near any equator, so the integral of a Lipschitz function over the sphere is asymptotically its value at a typical point (Convex – 9); this is the geometric counterpart of Laplace’s method.

1. Prove Laplace's method in $\mathbb{R}^n$ in the form of Theorem 1, and use it to obtain Stirling's formula from $n! = \int_0^\infty t^n e^{-t} dt$ and the asymptotics of $\int_{[0,1]^2} e^{-n(x^2+xy+y^2)} dx dy$.
2. (Proposed problems) Let $f: [0, 1]^2 \rightarrow \mathbb{R}$ be such that $\partial_x f$ and $\partial_y f$ are continuous. Find the value of the limit

$$\lim_{n \rightarrow \infty} n \left(n^2 \iint_{[0,1]^2} x^n y^n f(x, y) dx dy - f(1, 1) \right).$$

3. (Proposed problems) Prove that there exists n_0 such that for all $n \geq n_0$ the equation $n^2 \iint_{[0,1]^2} x^n y^n e^{xy^2} dx dy = e$ has a unique real solution z_n , and moreover $\lim_{n \rightarrow \infty} n(z_n - 1) = 4$.
4. (VJIMC 2005) Let $f: [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ be continuous. Find

$$\lim_{n \rightarrow \infty} \left(\frac{(2n+1)!}{(n!)^2} \right)^2 \int_0^1 \int_0^1 (xy(1-x)(1-y))^n f(x, y) dx dy.$$

5. Prove that the measure on the unit sphere S^{n-1} concentrates near the equator: for a fixed $\varepsilon > 0$ the relative measure of $\{|x_1| \geq \varepsilon\}$ decays exponentially in n ; deduce that for a 1-Lipschitz f on the sphere the integral differs from the median by $O(1/\sqrt{n})$.